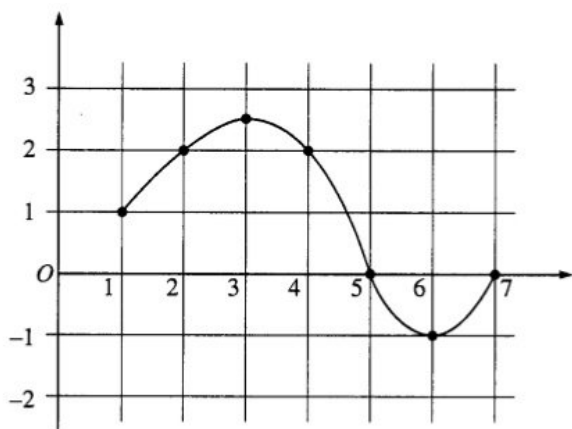


6-1

1.



The graph of a differentiable function f on the closed interval $[1, 7]$ is shown above. Let $h(x) = \int_1^x f(t) dt$ for $1 \leq x \leq 7$.

Find the value of x at which h has its minimum on the closed interval $[1, 7]$. Justify your answer.

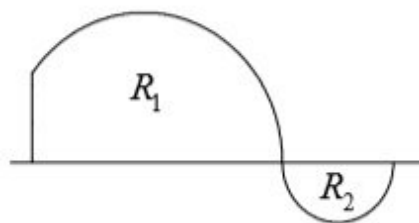
Part D

1 point is earned for the correct answer

1 point is earned for the correct justification where one point is earned for eliminating interior points and one point is earned for eliminating 7

minimum at $x = 1$ because:

h increases on $[1, 5]$ and decreases on $[5, 7]$, so minimum is at an endpoint



$$h(7) = \text{area } R_1 - \text{area } R_2 > 0 \text{ and } h(1) = 0$$



0	1	2	3
---	---	---	---

The student response earns one of the following points:

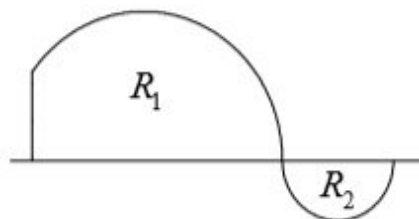
1 point is earned for the correct answer

6-1

2 points are earned for the correct justification where one point is earned for eliminating interior points and one point is earned for eliminating 7

minimum at $x = 1$ because:

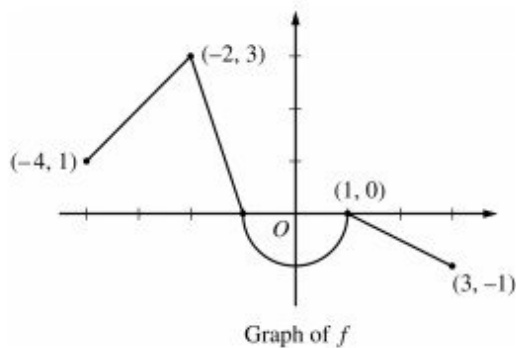
h increases on $[1, 5]$ and decreases on $[5, 7]$, so minimum is at an endpoint



$$h(7) = \text{area } R_1 - \text{area } R_2 > 0 \text{ and } h(1) = 0$$

6-1

2.



Let f be the continuous function defined on $[-4, 3]$ whose graph, consisting of three line segments and a semicircle centered at the origin, is given above. Let g be the function given by $g(x) = \int_1^x f(t) dt$.

For each of $g'(-3)$ and $g''(-3)$, find the value or state that it does not exist.

Part B

One point is earned for $g'(-3)$

One point is earned for $g''(-3)$

$$g'(x) = f(x) \Rightarrow g'(-3) = f(-3) = 2$$

$$g''(x) = f'(x) \Rightarrow g''(-3) = f'(-3) = 1$$



0	1	2
---	---	---

The student earns all of the following points:

One point is earned for $g'(-3)$

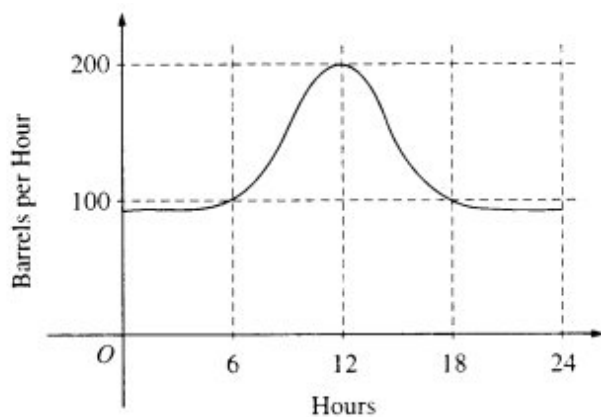
One point is earned for $g''(-3)$

$$g'(x) = f(x) \Rightarrow g'(-3) = f(-3) = 2$$

$$g''(x) = f'(x) \Rightarrow g''(-3) = f'(-3) = 1$$

6-1

3.



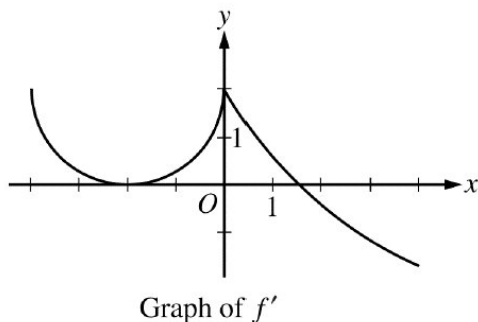
The flow of oil, in barrels per hour, through a pipeline on July 9 is given by the graph shown above. Of the following, which best approximates the total number of barrels of oil that passed through the pipeline that day?

- (A) 500
- (B) 600
- (C) 2,400
- (D) 3,000
- (E) 4,800



6-1

4.



The derivative of a function f is defined by $f'(x) = \begin{cases} g(x) & \text{for } -4 \leq x \leq 0 \\ 5e^{-x/3} - 3 & \text{for } 0 < x \leq 4 \end{cases}$.

The graph of the continuous function f , shown in the figure above, has x -intercepts at $x = -2$ and $x = 3 \ln\left(\frac{5}{3}\right)$. The graph of g on $-4 \leq x \leq 0$ is a semicircle, and $f(0) = 5$.

Find $f(-4)$ and $f(4)$.

Part B

1 point is earned for integral of $f(-4)$

1 point is earned for value of $f(-4)$

1 point is earned for integral of $f(4)$

1 point is earned for antiderivative of $f(4)$

1 point is earned for value of $f(4)$

$$f(-4) = 5 + \int_0^{-4} g(x) dx$$

$$= 5 - (8 - 2\pi) = 2\pi - 3$$

$$f(4) = 5 + \int_0^4 (5e^{-x/3} - 3) dx$$

$$= 5 + \left(-15e^{-x/3} - 3x\right) \Big|_{x=0}^{x=4}$$

$$= 8 - 15e^{-4/3}$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns five of the following points:

6-1

1 point is earned for integral of $f(-4)$

1 point is earned for value of $f(-4)$

1 point is earned for integral of $f(4)$

1 point is earned for antiderivative of $f(4)$

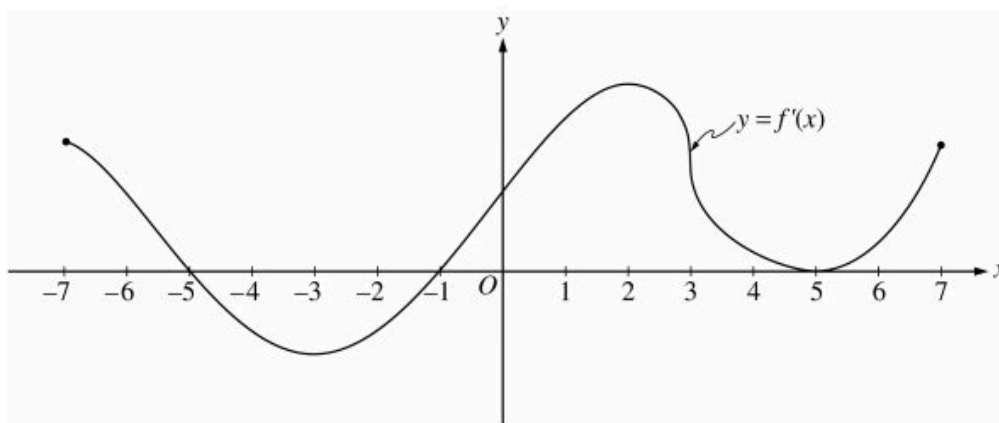
1 point is earned for value of $f(4)$

$$\begin{aligned}f(-4) &= 5 + \int_0^{-4} g(x) dx \\&= 5 - (8 - 2\pi) = 2\pi - 3\end{aligned}$$

$$\begin{aligned}f(4) &= 5 + \int_0^4 (5e^{\frac{-x}{3}} - 3) dx \\&= 5 + \left(-15e^{\frac{-x}{3}} - 3x\right) \Big|_{x=0}^{x=4} \\&= 8 - 15e^{\frac{-4}{3}}\end{aligned}$$

6-1

5.



The figure above shows the graph of f' , the derivative of the function f , for $-7 \leq x \leq 7$. The graph of f has horizontal tangent lines at $x = -3$, $x = 2$, and $x = 5$, and a vertical tangent line at $x = 3$.

- Find all values of x , for $-7 < x < 7$, at which f attains a relative minimum. Justify your answer.
- Find all values of x , for $-7 < x < 7$, at which f attains a relative maximum. Justify your answer.
- Find all values of x , for $-7 < x < 7$, at which $f''(x) < 0$.
- At what value of x , for $-7 \leq x \leq 7$, does f attain its absolute maximum? Justify your answer.

Part A

1 point is earned for the answer

1 point is earned for the justification

$x = -1$

$f'(x)$ changes from negative to positive at $x = -1$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

$x = -1$

$f'(x)$ changes from negative to positive at $x = -1$

Part B

1 point is earned for the answer

6-1

1 point is earned for the justification

$$x = -5$$

$f'(x)$ changes from positive to negative at $x = -5$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

$$x = -5$$

$f'(x)$ changes from positive to negative at $x = -5$

Part C

1 point is earned for $(-7, -3)$

1 point is earned for $(2, 3) \cup (3, 5)$

$f''(x)$ exists and f' is decreasing on the intervals $(-7, -3)$, $(2, 3)$, and $(3, 5)$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $(-7, -3)$

1 point is earned for $(2, 3) \cup (3, 5)$

$f''(x)$ exists and f' is decreasing on the intervals $(-7, -3)$, $(2, 3)$, and $(3, 5)$

Part D

1 point is earned for the answer

1 point is earned for identifying $x = -5$ and $x = 7$ as candidates

- or -

indicating that the graph of f increases, decreases, then increases

6-1

1 point is earned for justifying $f(7) > f(-5)$

$$x = 7$$

The absolute maximum must occur at $x = -5$ or at an endpoint.

$f(-5) > f(-7)$ because f is increasing on $(-7, -5)$

The graph of f' shows that the magnitude of the negative change in f from $x = -5$ to $x = -1$ is smaller than the positive change in f from $x = -1$ to $x = 7$. Therefore the net change in f is positive from $x = -5$ to $x = 7$, and $f(7) > f(-5)$. So $f(7)$ is the absolute maximum.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for identifying $x = -5$ and $x = 7$ as candidates

- or -

indicating that the graph of f increases, decreases, then increases

1 point is earned for justifying $f(7) > f(-5)$

$$x = 7$$

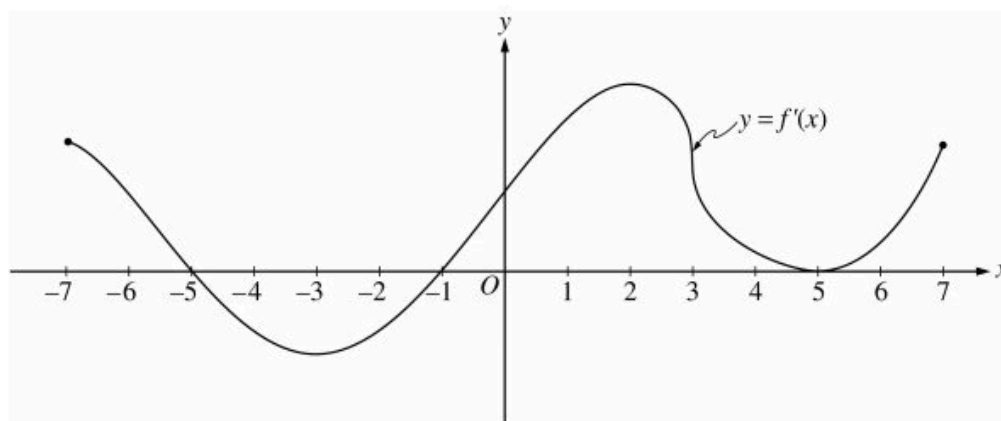
The absolute maximum must occur at $x = -5$ or at an endpoint.

$f(-5) > f(-7)$ because f is increasing on $(-7, -5)$

The graph of f' shows that the magnitude of the negative change in f from $x = -5$ to $x = -1$ is smaller than the positive change in f from $x = -1$ to $x = 7$. Therefore the net change in f is positive from $x = -5$ to $x = 7$, and $f(7) > f(-5)$. So $f(7)$ is the absolute maximum.

6-1

6.



The figure above shows the graph of f' , the derivative of the function f , for $-7 \leq x \leq 7$. The graph of f has horizontal tangent lines at $x = -3$, $x = 2$, and $x = 5$, and a vertical tangent line at $x = 3$.

At what value of x , for $-7 \leq x \leq 7$, does f attain its absolute maximum? Justify your answer.

Part D

1 point is earned for the answer

1 point is earned for identifies $x = -5$ and $x = 7$ as candidates

- or -

indicates that the graph of f increase, decreases, then increases

1 point is earned for justifies $f(7) > f(-5)$

$x = 7$

The absolute maximum must occur at $x = -5$ or at an endpoint.

$f(-5) > f(-7)$ because f is increasing on $(-7, -5)$

The graph of f' shows that the magnitude of the negative change in f from $x = -5$ to $x = -1$ is smaller than the positive change in f from $x = -1$ to $x = 7$. Therefore the net change in f is positive from $x = -5$ to $x = 7$, and $f(7) > f(-5)$. So $f(7)$ is the absolute maximum.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for identifies $x = -5$ and $x = 7$ as candidates

6-1

- or -

indicates that the graph of f increase, decreases, then increases

1 point is earned for justifies $f(7) > f(-5)$

$$x = 7$$

The absolute maximum must occur at $x = -5$ or at an endpoint.

$f(-5) > f(-7)$ because f is increasing on $(-7, -5)$

The graph of f' shows that the magnitude of the negative change in f from $x = -5$ to $x = -1$ is smaller than the positive change in f from $x = -1$ to $x = 7$. Therefore the net change in f is positive from $x = -5$ to $x = 7$, and $f(7) > f(-5)$. So $f(7)$ is the absolute maximum.

6-1

7. The function g is defined for $x > 0$ with $g(1)=2$, $g'(x)=\sin(x+1/x)$, and $g''(x)=(1-1/x^2)\cos(x+1/x)$.



Write an equation for the line tangent to the graph of g at $x=0.3$.

Part C

1 point is earned for $g'(0.3)$

1 point is earned for the integral expression

1 point is earned for $g(0.3)$

1 point is earned for the equation

$$g'(0.3) = -0.472161$$

$$g(0.3) = 2 + \int_1^{0.3} g'(x) dx = 1.546007$$

An equation for the line tangent to the graph of g is $y = 1.546 - 0.472(x - 0.3)$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $g'(0.3)$

1 point is earned for the integral expression

1 point is earned for $g(0.3)$

1 point is earned for the equation

$$g'(0.3) = -0.472161$$

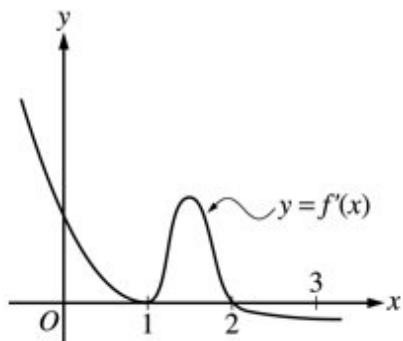
$$g(0.3) = 2 + \int_1^{0.3} g'(x) dx = 1.546007$$

An equation for the line tangent to the graph of g is

$$y = 1.546 - 0.472(x - 0.3).$$

6-1

8.



The graph of f' , the derivative of the function f , is shown above. If $f(0) = 0$, which of the following must be true?

I. $f(0) > f(1)$

II. $f(2) > f(1)$

III. $f(1) > f(3)$

(A) I only

(B) II only

(C) III only

(D) I and II only

(E) II and III only



6-1

The following are related to this scenario:

A store is having a 12-hour sale. The total number of shoppers who have entered the store t hours after the sale begins is modeled by the function S defined by $S(t) = 0.5t^4 - 16t^3 + 144t^2$ for $0 \leq t \leq 12$. At time $t = 0$, when the sale begins, there are no shoppers in the store.

9.  The rate at which shoppers leave the store, measured in shoppers per hour, is modeled by the function L defined by $L(t) = -80 + \frac{4400}{t^2 - 14t + 55}$ for $0 \leq t \leq 12$. According to the model, how many shoppers are in the store at the end of the sale (time $t = 12$)? Give your answers to the nearest whole number.

Part C

The response can earn up to 3 points:

1 point: For correct integral

1 point: For use of $S(12)$

1 point: For the correct answer

$$S(12) - \int_0^{12} L(t) dt = 195.701684$$

Therefore there are approximately 196 shoppers in the store at the end of the sale.



0	1	2	3
---	---	---	---

The response earns all three of the following points:

1 point: For correct integral

1 point: For use of $S(12)$

1 point: For the correct answer

$$S(12) - \int_0^{12} L(t) dt = 195.701684$$

Therefore there are approximately 196 shoppers in the store at the end of the sale.

6-1

10.  Find the value of $\frac{1}{3} \int_6^9 S'(t) dt$. Using correct units, explain the meaning of $\frac{1}{3} \int_6^9 S'(t) dt$ in the context of this problem.

Part B

The response can earn up to 2 points:

1 point: For the correct value of $\frac{1}{3} \int_6^9 S'(t) dt$

$$\begin{aligned} \frac{1}{3} \int_6^9 S'(t) dt &= \frac{1}{3}(S(9) - S(6)) \\ &= \frac{1}{3}(3280.5 - 2376) = 301.5 \end{aligned}$$

1 point: For the meaning of $\frac{1}{3} \int_6^9 S'(t) dt$

$\frac{1}{3} \int_6^9 S'(t) dt$ is the average rate of which the shoppers are entering the store in shoppers/hr between times $t = 6$ and $t = 9$ hours.



0	1	2
---	---	---

The response earns both of the following points:

1 point: For the correct value of $\frac{1}{3} \int_6^9 S'(t) dt$

$$\begin{aligned} \frac{1}{3} \int_6^9 S'(t) dt &= \frac{1}{3}(S(9) - S(6)) \\ &= \frac{1}{3}(3280.5 - 2376) = 301.5 \end{aligned}$$

1 point: For the meaning of $\frac{1}{3} \int_6^9 S'(t) dt$

6-1

$\frac{1}{3} \int_6^9 S'(t) dt$ is the average rate of which the shoppers are entering the store in shoppers/hr between times $t = 6$

and $t = 9$ hours.

11.  A cup of tea is cooling in a room that has a constant temperature of 70 degrees Fahrenheit ($^{\circ}\text{F}$). If the initial temperature of the tea, at time $t = 0$ minutes, is 200°F and the temperature of the tea changes at the rate $R(t) = -6.89e^{-0.053t}$ degrees Fahrenheit per minute, what is the temperature, to the nearest degree, of the tea after 4 minutes?

(A) 175°F



(B) 130°F

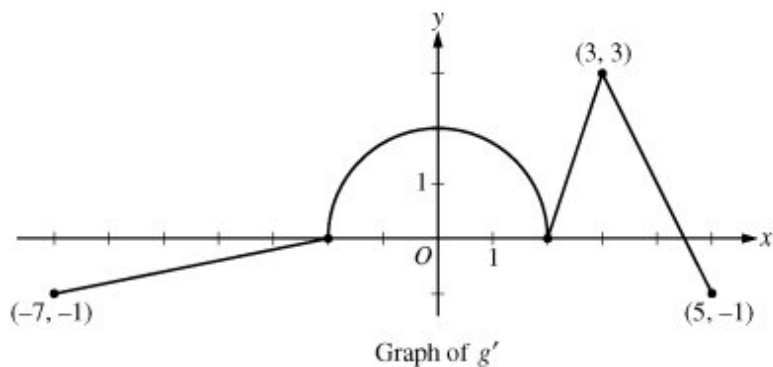
(C) 95°F

(D) 70°F

(E) 45°F

6-1

12.



The function g is defined and differentiable on the closed interval $[-7, 5]$ and satisfies $g(0) = 5$. The graph of $y = g'(x)$, the derivative of g , consists of a semicircle and three line segments, as shown in the figure above.

Find $g(3)$ and $g(-2)$.

Part A

The student response earns none of the following points:

1 point is earned for using $g(0)=5$

1 point is earned for $g(3)$

1 point is earned for $g(-2)$

$$g(3) = 5 + \int_0^3 g'(x) dx = 5 + \frac{\pi \cdot 2^2}{4} + \frac{3}{2} = \frac{13}{2} + \pi$$

$$g(-2) = 5 + \int_0^{-2} g'(x) dx = 5 - \pi$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for using $g(0)=5$

1 point is earned for $g(3)$

1 point is earned for $g(-2)$

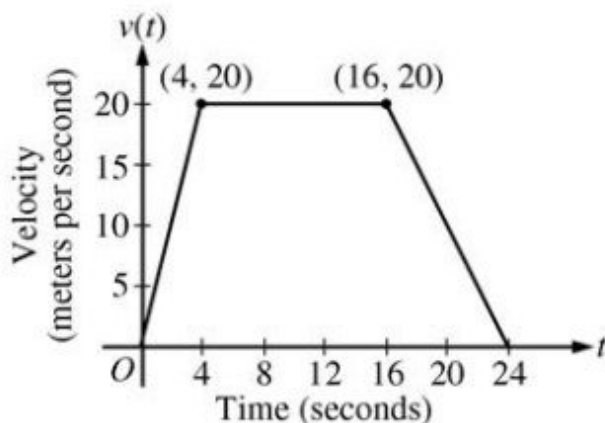
6-1

$$g(3) = 5 + \int_0^3 g'(x) dx = 5 + \frac{\pi \cdot 2^2}{4} + \frac{3}{2} = \frac{13}{2} + \pi$$

$$g(-2) = 5 + \int_0^{-2} g'(x) dx = 5 - \pi$$

6-1

13.



A car is traveling on a straight road. For $0 \leq t \leq 24$ seconds, the car's velocity $v(t)$, in meters per second, is modeled by the piecewise-linear function defined by the graph above.

Find $\int_0^{24} v(t) dt$. Using correct units, explain the meaning of $\int_0^{24} v(t) dt$.

Part A

1 point is earned for the value

1 point is earned for the meaning with units

$$\int_0^{24} v(t) dt = \frac{1}{2}(4)(20) + (12)(20) + \frac{1}{2}(8)(20) = 360$$

The car travels 360 meters in these 24 seconds.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the value

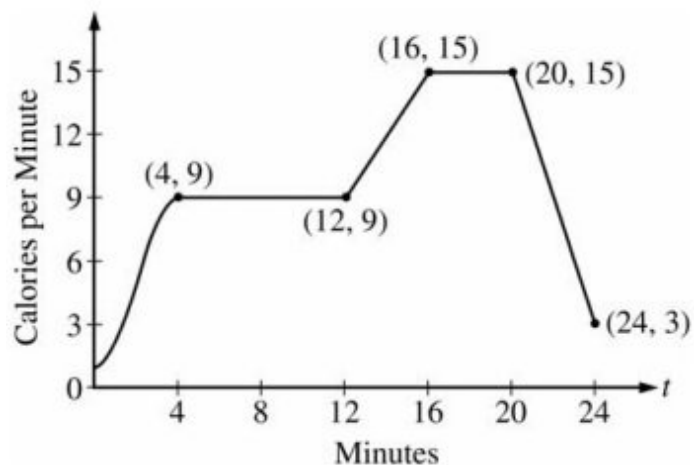
1 point is earned for the meaning with units

$$\int_0^{24} v(t) dt = \frac{1}{2}(4)(20) + (12)(20) + \frac{1}{2}(8)(20) = 360$$

The car travels 360 meters in these 24 seconds.

6-1

14.



The rate, in calories per minute, at which a person using an exercise machine burns calories is modeled by the function f . In the figure above, $f(t) = -\frac{1}{4}t^3 + \frac{3}{2}t^2 + 1$ for $0 \leq t \leq 4$ and f is piecewise linear for $4 \leq t \leq 24$.

Find the total number of calories burned over the time interval $6 \leq t \leq 18$ minutes.

Part C

1 point is earned for the method

1 point is earned for the answer

$$\int_0^{18} f(t) dt = 6(9) + \frac{1}{2}(4)(9 + 15) + 2(15) = 132 \text{ calories}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the method

1 point is earned for the answer

$$\int_0^{18} f(t) dt = 6(9) + \frac{1}{2}(4)(9 + 15) + 2(15) = 132 \text{ calories}$$

6-1

15.

t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W , where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t=0$, the temperature of the water is 55°F . The water is heated for 30 minutes, beginning at time $t=0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

Use the data in the table to evaluate $\int_0^{20} W'(t) dt$. Using correct units, interpret the meaning of $\int_0^{20} W'(t) dt$ in the context of this problem.

Part B

1 point is earned for the value

1 point is earned for the interpretation with units

$$\int_0^{20} W'(t) dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

The water has warmed by 16 oF over the interval from $t=0$ to $t=20$ minutes.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the value

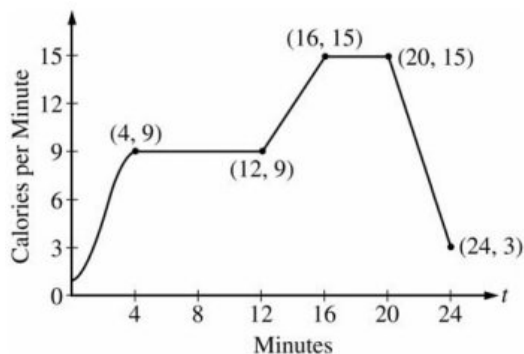
1 point is earned for the interpretation with units

$$\int_0^{20} W'(t) dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

The water has warmed by 16 OF over the interval from $t=0$ to $t=20$ minutes.

6-1

16.



The rate, in calories per minute, at which a person using an exercise machine burns calories is modeled by the function f . In the figure above, $f(t) = -\frac{1}{4}t^3 + \frac{3}{2}t^2 + 1$ for $0 \leq t \leq 4$ and f is piecewise linear for $4 \leq t \leq 24$.

- Find $f'(22)$. Indicate units of measure.
- For the time interval $0 \leq t \leq 24$, at what time t is f increasing at its greatest rate? Show the reasoning that supports your answer.
- Find the total number of calories burned over the time interval $6 \leq t \leq 18$ minutes.
- The setting on the machine is now changed so that the person burns $f(t) + c$ calories per minute. For this setting, find c so that an average of 15 calories per minute is burned during the time interval $6 \leq t \leq 18$.

Part A

1 point is earned for $f'(22)$ and units

$$f'(22) = \frac{15 - 3}{20 - 24} = -3 \text{ calories/min/min}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for $f'(22)$ and units

$$f'(22) = \frac{15 - 3}{20 - 24} = -3 \text{ calories/min/min}$$

Part B

1 point is earned for f' on $(0, 4)$

6-1

1 point is earned for showing f' has a max at $t = 2$ on $(0, 4)$

1 point is earned if shows $12 < t < 16, f'(t) < f'(2)$

1 point is earned for the answer

f is increasing on $[0, 4]$ and on $[12, 16]$.

On $(12, 16)$, $f'(t) = \frac{15 - 9}{16 - 12} = \frac{3}{2}$ since f has constant slope on this interval.

On $(0, 4)$, $f'(t) = -\frac{3}{4}t^2 + 3t$ and $f''(t) = -\frac{3}{2}t + 3 = 0$ when $t = 2$. This is where f' has a maximum on $[0, 4]$ since $f'' > 0$ on $(0, 2)$ and $f'' < 0$ on $(2, 4)$.

On $[0, 24]$, f is increasing at its greatest rate when $t = 2$ because $f'(2) = 3 > \frac{3}{2}$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for f' on $(0, 4)$

1 point is earned for showing f' has a max at $t = 2$ on $(0, 4)$

1 point is earned if shows $12 < t < 16, f'(t) < f'(2)$

1 point is earned for the answer

f is increasing on $[0, 4]$ and on $[12, 16]$.

On $(12, 16)$, $f'(t) = \frac{15 - 9}{16 - 12} = \frac{3}{2}$ since f has constant slope on this interval.

On $(0, 4)$, $f'(t) = -\frac{3}{4}t^2 + 3t$ and $f''(t) = -\frac{3}{2}t + 3 = 0$ when $t = 2$. This is where f' has a maximum on $[0, 4]$ since $f'' > 0$ on $(0, 2)$ and $f'' < 0$ on $(2, 4)$.

6-1

On $[0, 24]$, f is increasing at its greatest rate when $t = 2$ because $f'(2) = 3 > \frac{3}{2}$.

Part C

1 point is earned for the method

1 point is earned for the answer

$$\int_6^{18} f(t) dt = 6(9) + \frac{1}{2}(4)(9 + 15) + 2(15) = 132 \text{ calories}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the method

1 point is earned for the answer

$$\int_6^{18} f(t) dt = 6(9) + \frac{1}{2}(4)(9 + 15) + 2(15) = 132 \text{ calories}$$

Part D

1 point is earned for the setup

1 point is earned for the value of c

$$\text{We want } \frac{1}{12} \int_6^{18} (f(t) + c) dt = 15.$$

This means $132 + 12c = 15(12)$. So, $c = 4$.

OR

Currently, the average is $\frac{132}{12} = 11$ calories/min.

Adding c to $f(t)$ will shift the average by c . So $c = 4$ to get an average of 15 calories/min.

6-1



0	1	2
---	---	---

The student response earns none of the following points:

1 point is earned for the setup

1 point is earned for the value of c

We want $\frac{1}{12} \int_6^{18} (f(t) + c) dt = 15$.

This means $132 + 12c = 15(12)$. So, $c = 4$.

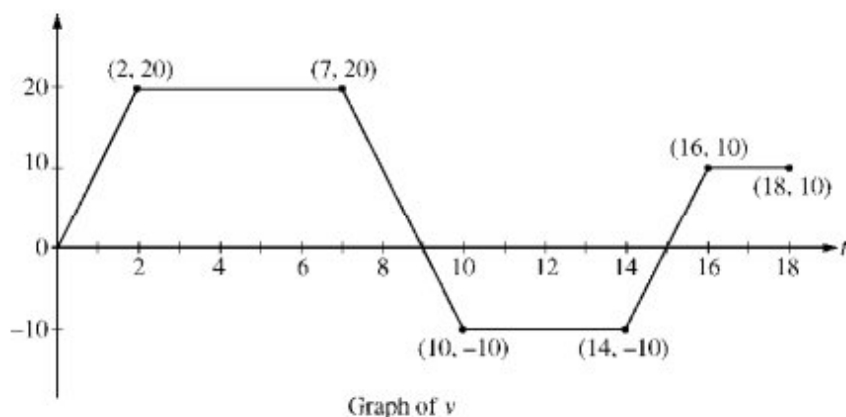
OR

Currently, the average is $\frac{132}{12} = 11$ calories/min.

Adding c to $f(t)$ will shift the average by c . So $c = 4$ to get an average of 15 calories/min.

6-1

17.



A squirrel starts at building A at time $t=0$ and travels along a straight, horizontal wire connected to building B . For $0 \leq t \leq 18$, the squirrel's velocity is modeled by the piecewise-linear function defined by the graph above.

At what time in the interval $0 \leq t \leq 18$ is the squirrel farthest from building A ? How far from building A is the squirrel at this time?

Part B

1 point is earned for identifying candidates

1 point is earned for answers

Velocity is 0 at $t=0$, $t=9$, and $t=15$.

t	position at time t
0	0
9	$\frac{9+5}{2} \cdot 20 = 140$
15	$140 - \frac{6+4}{2} \cdot 10 = 90$
18	$90 + \frac{3+2}{2} \cdot 10 = 115$

The squirrel is farthest from building A at time $t=9$;

its greatest distance from the building is 140.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for identifying candidates

6-1

1 point is earned for answers

Velocity is 0 at $t=0$, $t=9$, and $t=15$.

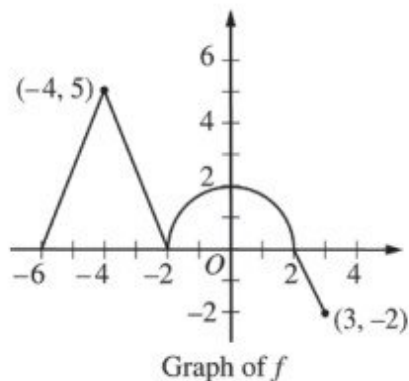
t	position at time t
0	0
9	$\frac{9+5}{2} \cdot 20 = 140$
15	$140 - \frac{6+4}{2} \cdot 10 = 90$
18	$90 + \frac{3+2}{2} \cdot 10 = 115$

The squirrel is farthest from building A at time $t=9$;

its greatest distance from the building is 140.

6-1

18.



The graph of the continuous function f , consisting of three line segments and a semicircle, is shown above.

Let g be the function given by $g(x) = \int_{-2}^x f(t) dt$.

Find $g(-6)$ and $g(3)$.

Part A

One point is earned for $g(-6)$

One point is earned for $g(3)$

$$g(-6) = \int_{-2}^{-6} f(t) dt = - \int_{-6}^{-2} f(t) dt = -\frac{1}{2} \cdot 4.5 = -10$$

$$g(3) = \int_{-2}^3 f(t) dt = \frac{1}{2} \pi \cdot 2^2 - \frac{1}{2} \cdot 1 \cdot 2 = 2\pi - 1$$



0	1	2
---	---	---

The student earns all of the following points:

One point is earned for $g(-6)$

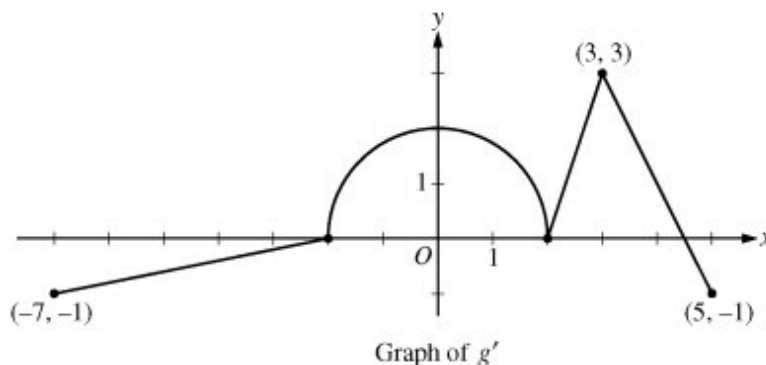
One point is earned for $g(3)$

$$g(-6) = \int_{-2}^{-6} f(t) dt = - \int_{-6}^{-2} f(t) dt = -\frac{1}{2} \cdot 4.5 = -10$$

$$g(3) = \int_{-2}^3 f(t) dt = \frac{1}{2} \pi \cdot 2^2 - \frac{1}{2} \cdot 1 \cdot 2 = 2\pi - 1$$

6-1

19.



The function g is defined and differentiable on the closed interval $[-7, 5]$ and satisfies $g(0) = 5$. The graph of $y = g'(x)$, the derivative of g , consists of a semicircle and three line segments, as shown in the figure above.

- (a) Find $g(3)$ and $g(-2)$.
- (b) Find the x -coordinate of each point of inflection of the graph of $y = g(x)$ on the interval $-7 < x < 5$. Explain your reasoning.
- (c) The function h is defined by $h(x) = g(x) - \frac{1}{2}x^2$. Find the x -coordinate of each critical point of h , where $-7 < x < 5$, and classify each critical point as the location of a relative minimum, relative maximum, or neither a minimum nor a maximum. Explain your reasoning.

Part A

1 point is earned for using $g(0) = 5$

1 point is earned for $g(3)$

1 point is earned for $g(-2)$

$$g(3) = 5 + \int_0^3 g'(x) dx = 5 + \frac{\pi \cdot 2^2}{2} + \frac{3}{2} = \frac{13}{2} + \pi$$

$$g(-2) = 5 + \int_0^{-2} g'(x) dx = 5 - \pi$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for using $g(0) = 5$

1 point is earned for $g(3)$

1 point is earned for $g(-2)$

6-1

$$g(3) = 5 + \int_0^3 g'(x)dx = 5 + \frac{\pi \cdot 2^2}{2} + \frac{3}{2} = \frac{13}{2} + \pi$$

$$g(-2) = 5 + \int_0^{-2} g'(x)dx = 5 - \pi$$

Part B

1 point is earned for identifying $x = 0, 2, 3$

1 point is earned for explanation

The graph of $y = g(x)$ has points of inflection at $x = 0, x = 2$, and $x = 3$ because g' changes from increasing to decreasing at $x = 0$ and $x = 3$, and g' changes from decreasing to increasing at $x = 2$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for identifying $x = 0, 2, 3$

1 point is earned for explanation

The graph of $y = g(x)$ has points of inflection at $x = 0, x = 2$, and $x = 3$ because g' changes from increasing to decreasing at $x = 0$ and $x = 3$, and g' changes from decreasing to increasing at $x = 2$.

Part C

1 point is earned for $h'(x)$

1 point is earned for identifying $x = \sqrt{2}, 3$

1 point is earned for answer for $\sqrt{2}$ with analysis

1 point is earned for answer for 3 with analysis

$$h'(x) = g'(x) - x = 0 \Rightarrow g'(x) = x$$

On the interval $-2 \leq x \leq 2$, $g'(x) = \sqrt{4 - x^2}$.

6-1

On this interval, $g'(x) = x$ when $x = \sqrt{2}$.

The only other solution to $g'(x) = x$ is $x = 3$.

$$h'(x) = g'(x) - x > 0 \text{ for } 0 \leq x < \sqrt{2}$$

$$h'(x) = g'(x) - x \leq 0 \text{ for } \sqrt{2} < x \leq 5$$

Therefore h has a relative maximum at $x = \sqrt{2}$, and h has neither a minimum nor a maximum at $x = 3$.



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for $h'(x)$

1 point is earned for identifying $x = \sqrt{2}$, 3

1 point is earned for answer for $\sqrt{2}$ with analysis

1 point is earned for answer for 3 with analysis

$$h'(x) = g'(x) - x = 0 \Rightarrow g'(x) = x$$

On the interval $-2 \leq x \leq 2$, $g'(x) = \sqrt{4 - x^2}$.

On this interval, $g'(x) = x$ when $x = \sqrt{2}$.

The only other solution to $g'(x) = x$ is $x = 3$.

$$h'(x) = g'(x) - x > 0 \text{ for } 0 \leq x < \sqrt{2}$$

$$h'(x) = g'(x) - x \leq 0 \text{ for } \sqrt{2} < x \leq 5$$

Therefore h has a relative maximum at $x = \sqrt{2}$, and h has neither a minimum nor a maximum at $x = 3$.

6-1

20. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	2	4	5	8	10	14
$h'(t)$ (feet per hour)	0.3	0.6	0.7	1.4	2.3	7.7

The height of water in a storage tank during a 15-hour period is given by a twice-differentiable function h , where $h(t)$ is measured in feet and t is measured in hours since midnight for $0 \leq t \leq 15$. The graph of h is concave up on the interval $0 \leq t \leq 15$. Selected values of the derivative of h , $h'(t)$, are given in the table above. At time $t = 2$, the height of the water is 7 feet.

(a) Use the tangent line approximation for h at time $t = 2$ to estimate $h(2.8)$, the height of the water at time $t = 2.8$. Is the approximation an overestimate or an underestimate for $h(2.8)$? Give a reason for your answer.

(b) Use a right Riemann sum with the five subintervals indicated by the data in the table to approximate $\int_2^{14} h'(t) \, dt$. Indicate units of measure.

(c) Is the approximation in part (b) an overestimate or underestimate for $\int_2^{14} h'(t) \, dt$? Give a reason for your answer.

(d) The sum $\sum_{k=1}^n h' \left(\frac{15}{n}(k-1) \right) \frac{15}{n}$ is a left Riemann sum with n subintervals of equal length. The limit of this sum as n goes to infinity can be interpreted as a definite integral. Express the limit as a definite integral.

Part A

For the first point, the approximation must show the form of a correct tangent line equation and evaluation of that equation at $t = 2.8$.

Numerical answers do not need to be simplified, and units are not required to earn the approximation point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-1



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ approximation
- ☐ underestimate with reason

Solution:

$$h(2.8) \approx h(2) + h'(2)(2.8 - 2) = 7 + 0.3(0.8) = 7.24 \text{ feet}$$

Since the graph of h is concave up on $2 \leq t \leq 2.8$, the tangent line lies under the graph of h on this interval. Therefore, the approximation is an underestimate for $h(2.8)$.

Part B

To earn the first point, the response must present the form of a right Riemann sum.

To earn the second point, values for h' must be substituted into the sum. Therefore, $2 \cdot 0.6 + 1 \cdot 0.7 + 3 \cdot 1.4 + 2 \cdot 2.3 + 4 \cdot 7.7$ earns the first and second points.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ right Riemann sum
- ☐ approximation
- ☐ units

Solution:

$$\begin{aligned} \int_2^{14} h'(t) dt &\approx (4 - 2)h'(4) + (5 - 4)h'(5) + (8 - 5)h'(8) + (10 - 8)h'(10) + (14 - 10)h'(14) \\ &= 2 \cdot 0.6 + 1 \cdot 0.7 + 3 \cdot 1.4 + 2 \cdot 2.3 + 4 \cdot 7.7 = 41.5 \text{ feet} \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

6-1



0	1
---	---

The student response accurately includes the criteria below.

- ☐ overestimate with reason that references the behavior of h'

Solution:

Since the graph of h is concave up on $0 \leq t \leq 15$, h' is increasing on $2 \leq t \leq 14$. Therefore, the right Riemann sum approximation is an overestimate for $\int_2^{14} h'(t) dt$.

Part D

The first point may be earned as presented in the solution or by using that information to determine that the interval of integration is of length 15. A response that produces the correct definite integral earns all 3 points.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ $\Delta t = \frac{15}{n}$
- ☐ limits of integration
- ☐ integrand

Solution:

For this Riemann sum, $\Delta t = \frac{15}{n}$.

Thus, the integral is over an interval of length 15. This interval can be taken to be $0 \leq t \leq 15$.

The left endpoints of the subintervals used in the Riemann sum would be of the form $t_k = \frac{15}{n}(k-1)$ for k from 1 to n .

$$\text{Therefore, } \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n h' \left(\frac{15}{n}(k-1) \right) \frac{15}{n} \right) = \int_0^{15} h'(t) dt.$$

6-1

Note: The limit of the Riemann sum can be written as any definite integral of the form $\int_a^{a+15} h'(t-a) dt$.

6-1

21. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	$\frac{1}{3}$	$\frac{2}{3}$	1
$R(t)$ (gallons per hour)	11	8	5	0

The rate at which water leaks from a container is modeled by the twice-differentiable function R , where $R(t)$ is measured in gallons per hour and t is measured in hours for $0 \leq t \leq 1$. Values of $R(t)$ are given in the table above for selected values of t .

- (a) Use the data in the table to find an approximation for $R'\left(\frac{1}{2}\right)$. Show the computations that lead to your answer. Indicate units of measure.
- (b) Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate $\int_0^1 R(t) \, dt$. Indicate units of measure.
- (c) Use the data in the table to evaluate $\int_0^{\frac{1}{3}} R'(t) \, dt$.
- (d) The sum $\sum_{k=1}^n R\left(\frac{1}{4} + \frac{k}{2n}\right) \frac{1}{2n}$ is a right Riemann sum with n subintervals of equal length. The limit of this sum as n goes to infinity can be interpreted as a definite integral. Express the limit as a definite integral.

Part A

Numerical answers do not need to be simplified for the approximation point. The approximation must include a substitution of the function values from the table into the difference quotient.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

6-1

The student response accurately includes both of the criteria below.

- ☐ approximation
- ☐ units

Solution:

$$R' \left(\frac{1}{2} \right) \approx \frac{R \left(\frac{2}{3} \right) - R \left(\frac{1}{3} \right)}{\frac{2}{3} - \frac{1}{3}} = \frac{5-8}{\frac{1}{3}} = -9 \text{ gallons per hour per hour}$$

Part B

To earn the first point, the response must present the form of a left Riemann sum. Therefore, $\frac{1}{3}(11 + 8 + 5)$ earns the first and second points.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ left Riemann sum
- ☐ approximation
- ☐ units

Solution:

$$\begin{aligned} \int_0^1 R(t) \, dt &\approx \frac{1}{3} \left(R(0) + R\left(\frac{1}{3}\right) + R\left(\frac{2}{3}\right) \right) \\ &= \frac{1}{3}(11 + 8 + 5) = 8 \text{ gallons} \end{aligned}$$

Part C

The response should include supporting work for the numerical answer of -3 .

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

6-1

The student response accurately includes a correct value.

Solution:

$$\int_0^{\frac{1}{3}} R'(t) dt = R\left(\frac{1}{3}\right) - R(0) = 8 - 11 = -3$$

Part D

The first point may be earned as presented in the solution or by using that information to determine that the interval of integration is of length $\frac{1}{2}$. A response that produces the correct definite integral earns all 3 points.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\Delta t = \frac{1}{2n}$
- ☐ limits of integration
- ☐ integrand

Solution:

For this Riemann sum, $\Delta t = \frac{1}{2n}$. Thus, the integral is over an interval of length $\frac{1}{2}$. This interval can be taken to be $\left[\frac{1}{4}, \frac{3}{4}\right]$.

The right endpoints of the subintervals used in the Riemann sum would be of the form $t_k = \frac{1}{4} + \frac{k}{2n}$ for k from 1 to n .

Therefore, $\lim_{n \rightarrow \infty} \left(\sum_{k=1}^n R\left(\frac{1}{4} + \frac{k}{2n}\right) \cdot \frac{1}{2n} \right) = \int_{\frac{1}{4}}^{\frac{3}{4}} R(t) dt.$

Note: The limit of the Riemann sum can be written as any definite integral of the form $\int_a^{a+\frac{1}{2}} R\left(\frac{1}{4} - a + t\right) dt.$

6-1

22. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	$\frac{1}{2}$	1	$\frac{3}{2}$	2
$R(t)$ (gallons per hour)	16	12	7	3	0

The rate at which oil leaks from a drum is modeled by the twice-differentiable function R , where $R(t)$ is measured in gallons per hour and t is measured in hours for $0 \leq t \leq 2$. Values of $R(t)$ are given in the table above for selected values of t .

(a) Use the data in the table to find an approximation for $R'\left(\frac{3}{4}\right)$. Show the computations that lead to your answer. Indicate units of measure.

(b) Use a trapezoidal sum with the four subintervals indicated by the data in the table to approximate $\int_0^2 R(t) \, dt$. Indicate units of measure.

(c) Use the data in the table to evaluate $\int_0^1 R'(t) \, dt$.

(d) The sum $\sum_{k=1}^n R\left(\frac{1}{2} + \frac{k}{n}\right) \frac{1}{n}$ is a right Riemann sum with n subintervals of equal length. The limit of this sum as n goes to infinity can be interpreted as a definite integral. Express the limit as a definite integral.

Part A

Numerical answers do not need to be simplified for the approximation point. The approximation must include a substitution of the function values from the table into the difference quotient.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

6-1

The student response accurately includes both of the criteria below.

- ☐ approximation
- ☐ units

Solution:

$$R' \left(\frac{3}{4} \right) \approx \frac{R(1) - R\left(\frac{1}{2}\right)}{1 - \frac{1}{2}} = \frac{7 - 12}{\frac{1}{2}} = -10$$

gallons per hour per hour

Part B

To earn the first point, the response must present the form of a trapezoidal sum. Therefore,

$\frac{1}{2} \left(\frac{1}{2}(16 + 12) + \frac{1}{2}(12 + 7) + \frac{1}{2}(7 + 3) + \frac{1}{2}(3 + 0) \right)$ earns the first and second points.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ trapezoidal sum
- ☐ approximation
- ☐ units

Solution:

$$\begin{aligned} \int_0^1 R(t) \, dt &\approx \frac{1}{2} \left(\frac{1}{2} \left(R(0) + R\left(\frac{1}{2}\right) \right) + \frac{1}{2} \left(R\left(\frac{1}{2}\right) + R(1) \right) + \frac{1}{2} \left(R(1) + R\left(\frac{3}{2}\right) \right) + \frac{1}{2} \left(R\left(\frac{3}{2}\right) + R(2) \right) \right) \\ &= \frac{1}{2} \left(\frac{1}{2} (16 + 12) + \frac{1}{2} (12 + 7) + \frac{1}{2} (7 + 3) + \frac{1}{2} (3 + 0) \right) \\ &= \frac{1}{4} (28 + 19 + 10 + 3) = 15 \text{ gallons} \end{aligned}$$

Part C

The response should include supporting work for the numerical answer of -9 .

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

6-1



0	1
---	---

The student response accurately includes a correct value.

Solution:

$$\int_0^1 R'(t) dt = R(1) - R(0) = 7 - 16 = -9$$

Part D

The first point may be earned as presented in the solution or by using that information to determine that the interval of integration is of length 1. A response that produces the correct definite integral earns all 3 points.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ $\Delta t = \frac{1}{n}$
- ☐ limits of integration
- ☐ integrand

Solution:

For this Riemann sum, $\Delta t = \frac{1}{n}$. Thus, the integral is over an interval of length 1. This interval can be taken to be $\left[\frac{1}{2}, \frac{3}{2}\right]$.

The right endpoints of the subintervals used in the Riemann sum would be of the form $t_k = \frac{1}{2} + \frac{k}{n}$ for k from 1 to n .

$$\text{Therefore, } \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n R\left(\frac{1}{2} + \frac{k}{n}\right) \cdot \frac{1}{n} \right) = \int_{\frac{1}{2}}^{\frac{3}{2}} R(t) dt.$$

Note: The limit of the Riemann sum can be written as any definite integral of the form $\int_a^{a+1} R\left(\frac{1}{2} - a + t\right) dt$.

6-1

23. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (weeks)	2	4	5	8	10
$L'(t)$ (feet per week)	1.0	0.8	0.7	0.4	0.2

The length of a vine during a 12-hour period is given by a twice-differentiable function L , where $L(t)$ is measured in feet and t is measured in weeks for $0 \leq t \leq 12$. The graph of L is concave down on the interval $0 \leq t \leq 12$. Selected values of the derivative of L , $L'(t)$, are given in the table above. At time $t = 4$, the length of the vine is 5 feet.

(a) Use the tangent line approximation for L at time $t = 4$ to estimate $L(4.3)$, the length of the vine at time $t = 4.3$. Is the approximation an overestimate or an underestimate for $L(4.3)$? Give a reason for your answer.

(b) Use a left Riemann sum with the four subintervals indicated by the data in the table to approximate $\int_2^{10} L'(t) \, dt$. Indicate units of measure.

(c) Is the approximation in part (b) an overestimate or an underestimate for $\int_2^{10} L'(t) \, dt$? Give a reason for your answer.

(d) The sum $\sum_{k=1}^n L'\left(\frac{10k}{n}\right) \frac{10}{n}$ is a right Riemann sum with n subintervals of equal length. The limit of this sum as n goes to infinity can be interpreted as a definite integral. Express the limit as a definite integral.

Part A

For the first point, the approximation must show the form of a correct tangent line equation and evaluation of that equation at $t = 4.3$. Numerical answers do not need to be simplified, and units are not required to earn the approximation point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-1



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ approximation
- ☐ underestimate with reason

Solution:

$$L(4.3) \approx L(4) + L'(4)(4.3 - 4) = 5 + 0.8(0.3) = 5.24 \text{ feet}$$

Since the graph of L is concave down on $4 \leq t \leq 4.3$, the tangent line lies above the graph of L on this interval. Therefore, the approximation is an overestimate for $L(4.3)$.

Part B

To earn the first point, the response must present the form of a left Riemann sum. To earn the second point, values for L' must be substituted into the sum. Therefore, $2 \cdot 1.0 + 1 \cdot 0.8 + 3 \cdot 0.7 + 2 \cdot 0.4$ earns the first and second points. Numerical answers do not need to be simplified for the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ left Riemann sum
- ☐ approximation
- ☐ units

Solution:

$$\begin{aligned} \int_2^{10} L'(t) dt &\approx (4 - 2)L'(2) + (5 - 4)L'(4) + (8 - 5)L'(5) + (10 - 8)L'(8) \\ &= 2 \cdot 1.0 + 1 \cdot 0.8 + 3 \cdot 0.7 + 2 \cdot 0.4 = 5.7 \text{ feet} \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

6-1



0	1
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The student response accurately includes overestimate with reason that references the behavior of L'

Solution:

Since the graph of L is concave down on $0 \leq t \leq 12$, L' is decreasing on $2 \leq t \leq 10$. Therefore, the left Riemann sum approximation is an overestimate for $\int_2^{10} L'(t) dt$.

Part D

The first point may be earned as presented in the solution or by using that information to determine that the interval of integration is of length 10. A response that produces the correct definite integral earns all 3 points.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ $\Delta t = \frac{10}{n}$
- ☐ limits of integration
- ☐ integrand

Solution:

For this Riemann sum, $\Delta t = \frac{10}{n}$. Thus, the integral is over an interval of length 10. This interval can be taken to be $0 \leq t \leq 10$.

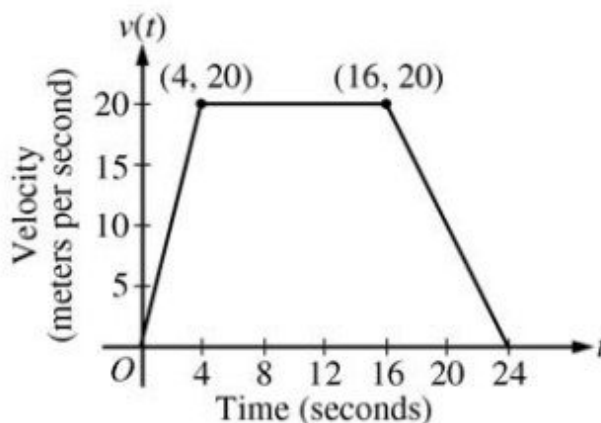
The right endpoints of the subintervals used in the Riemann sum would be of the form $t_k = \frac{10k}{n}$ for k from 1 to n .

Therefore, $\lim_{n \rightarrow \infty} \left(\sum_{k=1}^n L' \left(\frac{10k}{n} \right) \frac{10}{n} \right) = \int_0^{10} L'(t) dt$.

Note: The limit of the Riemann sum can be written as any definite integral of the form $\int_a^{a+10} L'(t-a) dt$.

6-1

24.



A car is traveling on a straight road. For $0 \leq t \leq 24$ seconds, the car's velocity $v(t)$, in meters per second, is modeled by the piecewise-linear function defined by the graph above.

- (a) Find $\int_0^{24} v(t) dt$. Using correct units, explain the meaning of $\int_0^{24} v(t) dt$.
- (b) For each of $v'(4)$ and $v'(20)$, find the value or explain why it does not exist. Indicate units of measure.
- (c) Let $a(t)$ be the car's acceleration at time t , in meters per second per second. For $0 < t < 24$, write a piecewise-defined function for $a(t)$.
- (d) Find the average rate of change of v over the interval $8 \leq t \leq 20$. Does the Mean Value Theorem guarantee a value of c , for $8 < c < 20$, such that $v'(c)$ is equal to this average rate of change? Why or why not?

Part A

1 point is earned for the value

1 point is earned for the meaning with units

$$\int_0^{24} v(t) dt = \frac{1}{2}(4)(20) + (12)(20) + \frac{1}{2}(8)(20) = 360$$

The car travels 360 meters in these 24 seconds.



0	1	2
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The student response earns all of the following points:

1 point is earned for the value

1 point is earned for the meaning with units

6-1

$$\int_0^{24} v(t) dt = \frac{1}{2}(4)(20) + (12)(20) + \frac{1}{2}(8)(20) = 360$$

The car travels 360 meters in these 24 seconds.

Part B

1 point is earned for: $v'(4)$ does not exist, with explanation

1 point is earned for: $v'(20)$

1 point is earned for the units

$v'(4)$ does not exist because

$$\lim_{t \rightarrow 4^-} \left(\frac{v(t) - v(4)}{t - 4} \right) = 5 \neq 0 = \lim_{t \rightarrow 4^+} \left(\frac{v(t) - v(4)}{t - 4} \right)$$

$$v'(20) = \frac{20 - 0}{16 - 24} = -\frac{5}{2} \text{ m/sec}^2$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for: $v'(4)$ does not exist, with explanation

1 point is earned for: $v'(20)$

1 point is earned for the units

$v'(4)$ does not exist because

$$\lim_{t \rightarrow 4^-} \left(\frac{v(t) - v(4)}{t - 4} \right) = 5 \neq 0 = \lim_{t \rightarrow 4^+} \left(\frac{v(t) - v(4)}{t - 4} \right)$$

$$v'(20) = \frac{20 - 0}{16 - 24} = -\frac{5}{2} \text{ m/sec}^2$$

Part C

1 point is earned for finds the values 5, 0, $-\frac{5}{2}$

1 point is earned for identifies constants with correct intervals

6-1

$$a(t) = \begin{cases} 5 & \text{if } 0 < t < 4 \\ 0 & \text{if } 4 < t < 16 \\ -\frac{5}{2} & \text{if } 16 < t < 24 \end{cases}$$

$a(t)$ does not exist at $t = 4$ and $t = 16$.



0	1	2
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The student response earns all of the following points:

1 point is earned for finds the values 5 , 0 , $-\frac{5}{2}$

1 point is earned for identifies constants with correct intervals

$$a(t) = \begin{cases} 5 & \text{if } 0 < t < 4 \\ 0 & \text{if } 4 < t < 16 \\ -\frac{5}{2} & \text{if } 16 < t < 24 \end{cases}$$

$a(t)$ does not exist at $t = 4$ and $t = 16$.

Part D

1 point is earned for the average rate of change of v on $[8, 20]$

1 point is earned for the answer with explanation

The average rate of change of v on $[8, 20]$ is

$$\frac{v(20) - v(8)}{20 - 8} = -\frac{5}{6} \text{ m/sec}^2.$$

No, the Mean Value Theorem does not apply to v on $[8, 20]$ because v is not differentiable at

$t = 16$.



0	1	2
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The student response earns all of the following points:

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6-1

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$$\frac{v(20) - v(8)}{20 - 8} = -\frac{5}{6} \text{ m/sec}^2.$$

No, the Mean Value Theorem does not apply to v on $[8, 20]$ because v is not differentiable at

$t = 16$.

6-1

25.



t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W , where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t = 0$, the temperature of the water is 55°F . The water is heated for 30 minutes, beginning at time $t = 0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

(a) Use the data in the table to estimate $W'(12)$. Show the computations that lead to your answer. Using correct units, interpret the meaning of your answer in the context of this problem.

(b) Use the data in the table to evaluate $\int_0^{20} W'(t) dt$. Using correct units, interpret the meaning of $\int_0^{20} W'(t) dt$ in the context of this problem.

(c) For $0 \leq t \leq 20$, the average temperature of the water in the tub is $\frac{1}{20} \int_0^{20} W(t) dt$. Use a left Riemann sum with the four subintervals indicated by the data in the table to approximate $\frac{1}{20} \int_0^{20} W(t) dt$. Does this approximation overestimate or underestimate the average temperature of the water over these 20 minutes? Explain your reasoning.

(d) For $20 \leq t \leq 25$, the function W that models the water temperature has first derivative given by $W'(t) = 0.4\sqrt{t} \cos(0.06t)$. Based on the model, what is the temperature of the water at time $t = 25$?

Part A

1 point is earned for the estimation

1 point is earned for the interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6}$$

$$= 1.017 \text{ (or } 1.016)$$

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.



0	1	2
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The student response earns all of the following points:

1 point is earned for the estimation

1 point is earned for the interpretation with units

6-1

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6}$$

$$= 1.017 \text{ (or } 1.016)$$

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.

Part B

1 point is earned for the value

1 point is earned for the interpretation with units

$$\int_0^{20} W'(t) dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.



0	1	2
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The student response earns all of the following points:

1 point is earned for the value

1 point is earned for the interpretation with units

$$\int_0^{20} W'(t) dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.

Part C

1 point is earned for the left Riemann sum

1 point is earned for the approximation

1 point is earned for underestimate with reason

6-1

$$\begin{aligned}\frac{1}{20} \int_0^{20} W(t) dt &\approx \frac{1}{20}(4 \cdot W(0) + 5 \cdot W(4) + 6 \cdot W(9) + 5 \cdot W(15)) \\ &= \frac{1}{20}(4 \cdot 55.0 + 5 \cdot 57.1 + 6 \cdot 61.8 + 5 \cdot 67.9) \\ &= \frac{1}{20} \cdot 1215.8 = 60.79\end{aligned}$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the left Riemann sum

1 point is earned for the approximation

1 point is earned for underestimate with reason

$$\begin{aligned}\frac{1}{20} \int_0^{20} W(t) dt &\approx \frac{1}{20}(4 \cdot W(0) + 5 \cdot W(4) + 6 \cdot W(9) + 5 \cdot W(15)) \\ &= \frac{1}{20}(4 \cdot 55.0 + 5 \cdot 57.1 + 6 \cdot 61.8 + 5 \cdot 67.9) \\ &= \frac{1}{20} \cdot 1215.8 = 60.79\end{aligned}$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing.

Part D

1 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned}W(25) &= 71.0 + \int_{20}^{25} W'(t) dt \\ &= 71.0 + 2.043155 = 73.043\end{aligned}$$

6-1



0	1	2
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The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the answer

$$W(25) = 71.0 + \int_{20}^{25} W'(t) dt$$

$$= 71.0 + 2.043155 = 73.043$$